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Homework 2: Circuit Emulation (Four Ways)

CIRCUIT DESCRIPTION

`Emulate a circuit that takes in 4 inputs (A, B, C, D) and whose output Q is True only when` **`exactly`** `1 of the 4 inputs is` **`False.`**

Introduction

I started with the truth table and easily derived a Boolean expression from it. After the boolean expression, I worked on the circuit diagram, then the Python function. Finishing the assignment off with the optional extra credit simplification. The truth table was the easiest for me. I now have a solid understanding of how to make them. The most challenging was the circuit diagram. I get lost about halfway through and start crossing wires or putting them in the incorrect gate. I find myself starting over multiple times. This time, I labeled the gates as I went, which helped, but I still found I needed more space than I had initially thought for my gates and wires. I regret not spreading out my inputs and gates, which made it hard to keep everything neat as I neared the end.

Truth Table/Test Table

A	**B**	**C**	**D**	**Q (expected)**	**Rc (actual)**	**Rp (actual)**	**Subexpression**
0	0	0	0	0	0	0	
0	0	0	1	0	0	0	
0	0	1	0	0	0	0	
0	0	1	1	0	0	0	
0	1	0	0	0	0	0	
0	1	0	1	0	0	0	
0	1	1	0	0	0	0	
0	1	1	1	1	1	1	~A ^ B^ C ^ D
1	0	0	0	0	0	0	
1	0	0	1	0	0	0	
1	0	1	0	0	0	0	
1	0	1	1	1	1	1	A ^ ~B^ C ^ D
1	1	0	0	0	0	0	

1	1	0	1	1	1	1	A ^ B ^ ~C ^ D
1	1	1	0	1	1	1	A ^ B^ C ^ ~D
1	1	1	1	0	0	0	

Boolean Algebra Expression

(~A ^ B^ C ^ D) v (A ^ ~B^ C ^ D) v (A ^ B ^ ~C ^ D) v (A ^ B^ C ^ ~D)

[Optional for extra credit] Simplification

Step 1. Start with (~A ^ B^ C ^ D) v (A ^ ~B^ C ^ D) v (A ^ B ^ ~C ^ D) v (A ^ B^ C ^ ~D)

Step 2. Group terms using the associative law

((~A ^ B^ C ^ D) v (A ^ ~B^ C ^ D)) v ((A ^ B ^ ~C ^ D) v (A ^ B^ C ^ ~D))

Step 3 Factor the like terms in the first group using the distributive law

(C ^ D ^ ((~A ^ B) v (A ^ ~B)))

Step 4 Factor the like terms second group using the distributive law

(A ^ B ^ ((~C ^ D) v (C ^ ~D)))

Step 5 Combine the expressions

(C ^ D ^ ((~A ^ B) v (A ^ ~B))) v (A ^ B ^ ((~C ^ D) v (C ^ ~D)))

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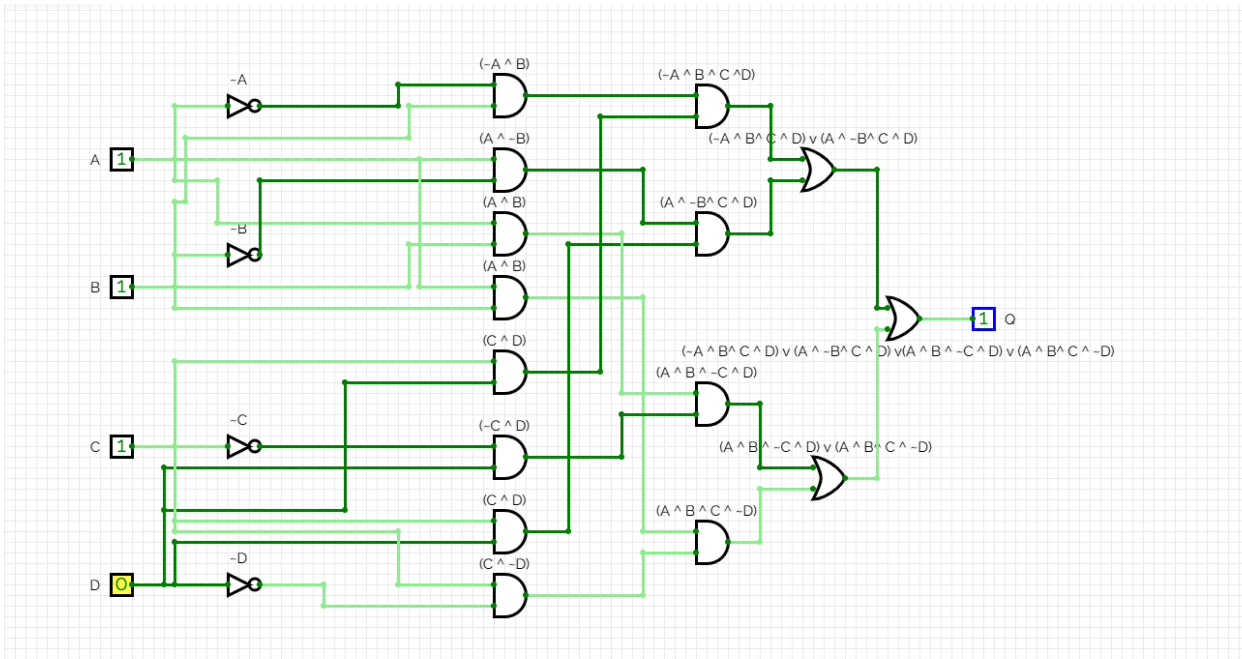
Circuit Diagram



Python Function

```
def circuit(a, b, c, d):
    """emulates (~A ^ B ^ C ^ D) v (A ^ ~B ^ C ^ D) v (A ^ B ^ ~C ^ D) v (A ^ B ^ C ^ ~D)"""
    gate1 = gate_not(a)  # ~A
    gate2 = gate_and(gate1, b)  # ~A ^ B
    gate3 = gate_and(gate2, c)  # ~A ^ B ^ C
    gate4 = gate_and(gate3, d)  # ~A ^ B ^ C ^ D (Term 1)

    gate5 = gate_not(b)  # ~B
    gate6 = gate_and(a, gate5)  # A ^ ~B
    gate7 = gate_and(gate6, c)  # A ^ B ^ C
    gate8 = gate_and(gate7, d)  # A ^ B ^ C ^ D (Term 2)

    gate9 = gate_not(c)  # ~C
    gate10 = gate_and(a, b)  # A ^ B
    gate11 = gate_and(gate10, gate9)  # A ^ B ^ ~C
    gate12 = gate_and(gate11, d)  # A ^ B ^ ~C ^ D (Term 3)

    gate13 = gate_not(d)  # ~D
    gate14 = gate_and(a, b)  # A ^ B
    gate15 = gate_and(gate14, c)  # A ^ B ^ C
    gate16 = gate_and(gate15, gate13)  # A ^ B ^ C ^ ~D

    gate17 = gate_or(gate4, gate8)  # Term1 V Term2
    gate18 = gate_or(gate12, gate16)  # Term3 V Term4
    gate19 = gate_or(gate17, gate18)  # (Term1 V Term2) V (Term3 V Term4)
```

```
    return gate19


print(circuit(0, 1, 1, 1))
```